DS220: Data Mining and warehousing I

Rational

In the modern data-driven world, organizations rely on data mining and warehousing to extract valuable insights from vast datasets. These technologies are critical for making data-driven decisions, uncovering hidden trends, and improving business processes. This course is designed to provide students with a strong foundation in these fields, enabling them to address the growing demand for professionals with expertise in data warehousing, data mining, and business intelligence.

Learning outcomes

By the end of the course-tenure the students should be able to:

- Design a data warehouse system and perform business analysis with OLAP tools.
- Apply suitable data pre-processing and data visualization techniques for data analysis.
- Apply frequent pattern and association rule mining techniques for data analysis.
- Apply appropriate classification techniques for data analysis.
- Apply appropriate clustering techniques for data analysis.
- Understand the fundamental concepts of data warehousing and its advantages.
- Explain the differences between OLTP systems and data warehouses.
- Utilize top-down and bottom-up development methodologies in data warehousing projects.

Course content

Unit 1: Data Warehouse Fundamentals

- ❖ Introduction to Data Warehouse and its role in modern data management.
- ❖ Distinct characteristics and advantages of Data Warehouses compared to OLTP systems.
- ❖ Overview of top-down and bottom-up development methodologies.
- ❖ Understanding different types of Data Warehouses.
- ❖ Tools and technologies used in Data Warehouse development.
- ❖ Historical development and evolution of Data Warehouses.

- Business intelligence trends and their influence on Data Warehousing.
- The impact of cloud computing on Data Warehouse architecture.
- Scalability and performance considerations in Data Warehousing.
- Case studies of successful Data Warehouse implementations.

Unit 2: Planning and Requirements

- The importance of planning and key considerations in Data Warehouse projects.
- Project management strategies for constructing Data Warehouses.
- Data Warehouse project life cycle and the Kimball Lifecycle Diagram.
- Approaches to requirements gathering, including team organization and roles.
- Best practices in stakeholder communication and managing expectations.
- Risk assessment and mitigation strategies in Data Warehouse planning.
- Real-world examples of challenges and solutions in Data Warehouse projects.
- The role of data governance and data quality in project planning.
- Business alignment and strategic alignment in Data Warehouse planning.
- Managing changing requirements and scope in Data Warehouse projects.

Unit 3: Data Warehouse Architecture

- Components and technical aspects of Data Warehouse architecture.
- Exploration of various Data Warehouse architectures, including federated Data Warehouses.
- Tool selection and its significance in Data Warehouse architecture.
- Scalability and elasticity considerations in Data Warehouse architecture.
- Security and access control in Data Warehouse architecture.
- Data integration and data modeling in Data Warehouse architecture.
- Data warehousing in the context of hybrid and multi-cloud environments.
- Case studies of organizations with unique Data Warehouse architectures.
- Performance optimization techniques in Data Warehouse architecture.
- Data Warehouse disaster recovery and business continuity planning.

Unit 4: Dimensional Modeling

- Introduction to Dimensional Modeling and its importance in Data Warehousing.
- Comparison between E-R Modeling and Dimensional Modeling.

- In-depth examination of Data Warehouse schemas, such as Star Schema, Snowflake Schema, and Fact Constellation Schema.
- Modeling hierarchies and slowly changing dimensions in Data Warehousing.
- Advanced dimensional modeling techniques for handling complex business scenarios.
- Real-world examples of effective dimensional modeling in Data Warehouses.
- Data modeling for unstructured and semi-structured data in Data Warehousing.
- The role of metadata and data dictionaries in dimensional modeling.
- Implementing self-service analytics through well-designed dimensional models.
- Scalability and performance considerations in Dimensional Modeling.

Unit 5: Extract, Transform, and Load (ETL)

- Overview of the ETL process, its requirements, and key steps.
- Data extraction methods, including logical and physical extraction.
- Data transformation techniques and various transformation types.
- Data loading methods and the role of ETL tools in the process.
- Data profiling and data quality assessment in ETL processes.
- Handling real-time and streaming data in ETL workflows.
- ETL best practices for data integration and data consolidation.
- Data lineage and data auditing in ETL processes.
- ETL automation and orchestration using modern tools.
- Case studies of successful ETL implementations in Data Warehousing.

Unit 6: Data Warehouse and OLAP

- Explanation of OLAP (Online Analytical Processing) and its characteristics.
- The OLAP creation process and its advantages.
- Multidimensional data in the context of Data Warehousing.
- Understanding different OLAP architectures, including MOLAP, ROLAP, and HOLAP.
- In-memory OLAP and columnar databases in modern Data Warehousing.
- Real-time OLAP and interactive data exploration techniques.
- Advanced OLAP features, such as drill-through and what-if analysis.
- Integration of OLAP with data visualization and reporting tools.
- Mobile OLAP and accessibility for decision-makers on the go.

- OLAP performance tuning and query optimization strategies.

Unit 7: Metadata Management in Data Warehouse

- Introduction to metadata and its categorization.
- Practical approaches to metadata management, including requirements gathering and classification.
- Use of metadata management tools, with a focus on Oracle and SAS solutions.
- Metadata repositories and data lineage tracking in Data Warehousing.
- Data cataloging and data dictionary management for metadata.
- Impact of metadata on data discovery and self-service analytics.
- Metadata-driven automation and data governance in Data Warehousing.
- Real-world examples of organizations leveraging metadata for business insights.
- The role of metadata in compliance, audit, and data security.
- Data lineage visualization and its importance in metadata management.

Unit VIII: Data Mining and Business Intelligence

- Overview of data mining and its scope.
- Understanding predictive modeling and its role in data mining.
- Integration of data mining with data warehousing.
- Profitable applications and tools for data mining.
- Real-time and streaming data mining for instant insights.
- The impact of artificial intelligence and machine learning on data mining.
- Advanced data visualization techniques for data mining results.
- Data storytelling and communicating data mining insights effectively.
- Case studies of successful data mining projects in various industries.
- Ethical considerations and privacy in data mining and business intelligence.

Unit IX: Business Intelligence

- Introduction to Business Intelligence (BI) and its significance.
- Business Intelligence tools and infrastructure.
- Real-world applications of Business Intelligence and its differentiation from Data Warehousing and Data Mining.
- The future of Business Intelligence.
- Self-service BI and data democratization in organizations.

- ❖ Embedded BI and analytics in enterprise applications.
- ❖ Advanced BI features, including natural language querying and augmented analytics.
- ❖ BI in the cloud and hybrid BI architectures.
- ❖ BI governance and data-driven decision-making in organizations.
- ❖ Business Intelligence in small and medium-sized enterprises (SMEs).

Unit X: Data Pre-processing

- ❖ Introduction to data pre-processing and its importance.
- ❖ Overview of data cleaning, integration, transformation, reduction, and discretization.
- ❖ Techniques for concept hierarchy generation.
- ❖ Handling missing data and imputation techniques.
- ❖ Data standardization and normalization for analysis.
- ❖ Exploratory data analysis (EDA) and data profiling.
- ❖ Data pre-processing for unstructured and semi-structured data.
- ❖ Real-world challenges in data preprocessing and their solutions.
- ❖ Automation of data pre-processing tasks using machine learning.
- ❖ Data pre-processing in big data environments and distributed data processing.

E 1.8.4 Teaching and Learning Methods

Lectures:90Hrs Tutorials:60Hrs, Labs:60Hrs, Assessments:30Hrs, NATIONAL HRS: 240

E1.8.4 Assessment

- Continuous Assessment 40%
  - 20% for hands-on assessments
  - 20% for non-hands-on assessments
- Final Exams 60%
  - 60% for theory assessment or
  - 40% theory assessment and 20% practical assessment

E1.8.5 Recommended Readings

- Berson, A., & Smith, S. J. (2016). Data Warehousing, Data Mining & OLAP (35th Reprint). Tata McGraw-Hill Edition.
- Soman, K. P., Diwakar, S., & Ajay, V. (2006). Insight into Data Mining: Theory and Practice. Eastern Economy Edition, Prentice Hall of India.

- Witten, I. H., & Frank, E. (2016). Data Mining: Practical Machine Learning Tools and Techniques (Second Edition). Elsevier.
- Han, J., Kamber, M., & Pei, J. (2012). Data Mining: Concepts and Techniques. Morgan Kaufmann.